
DISCRETE STRUCTURES I

Predicates and Quantifiers

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1 Introduction

Propositional logic, studied previously, cannot adequately express the meaning of all statements in mathematics and in natural language. For example, suppose that we know that "Every computer connected to the university network is functioning properly."

No rules of propositional logic allow us to conclude the truth of the statement "MATH3 is functioning properly," where MATH3 is one of the computers connected to the university network. Likewise, we cannot use the rules of propositional logic to conclude from the statement "CS2 is under attack by an intruder" where CS2 is a computer on the university network, to conclude the truth of "There is a computer on the university network that is under attack by an intruder."

In this section we will introduce a more powerful type of logic called predicate logic. We will see how predicate logic can be used to express the meaning of a wide range of statements in mathematics and computer science in ways that permit us to reason and explore relationships between objects.

2 Predicates

"Statements" in Logic of predicate is an "expression" involving variables such as "computer x is under attack by an intruder", " $x > 3$ ", or " $x = y + 3$ ".

The statement " x is greater than 3" has two parts :

1. First part : the variable x , is the subject of the statement.
2. Second part : "is greater than 3" refers to the property that the subject x has.

Definition 1. *A predicate refers to a property that a subject of a statement can have.*

We can denote the statement " x is greater than 3" by $P(x)$, where P denotes the predicate "is greater than 3" and x is the variable.

The statement $P(x)$ is also said to be the value of the propositional function P at x . Once a value has been assigned to the variable x , the statement $P(x)$ becomes a proposition and has a truth value.

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Example 1 : Let $P(x)$ denote the statement $x > 3$. What are the truth values of $P(4)$ and $P(2)$?

We obtain the statement $P(4)$ by setting $x = 4$ in the statement $x > 3$. Hence, $P(4)$, which is the statement $4 > 3$ is true. However, $P(2)$, which is the statement $2 > 3$ is false.

We can also have statements that involve more than one variable. For instance, consider the statement $x = y + 3$. We can denote this statement by $Q(x, y)$, where x and y are variables and Q is the predicate. When values are assigned to the variables x and y , the statement $Q(x, y)$ has a truth value.

Example 2 : Let $Q(x, y)$ denote the statement $x = y + 3$. What are the truth values of the propositions $Q(1, 2)$ and $Q(3, 0)$?

To obtain $Q(1, 2)$, set $x = 1$ and $y = 2$ in the statement $Q(x, y)$. Hence, $Q(1, 2)$ is the statement $1 = 2 + 3$ which is false. The statement $Q(3, 0)$ is the proposition $3 = 0 + 3$ which is true.

Example 3 : Let $R(x, y, z)$ denote the statement " $x + y = z$ ". What are the truth values of the propositions $R(1, 2, 3)$ and $R(0, 0, 1)$?

In general, a statement involving the n variables $x_1, x_2, \dots, x_n$ can be denoted by $P(x_1, x_2, \dots, x_n)$.

Remark : Propositional functions occur in computer programs :

Example :

Consider the statement if $x > 0$ then $x := x + 1$.

When this statement is encountered in a program, the value of the variable x at that point in the execution of the program is inserted into $P(x)$, which is " $x > 0$ ". If $P(x)$ is true for this value of x , the assignment statement $x := x + 1$ is executed, so the value of x is increased by 1. If $P(x)$ is false for this value of x , the assignment statement is not executed, so the value of x is not changed.

3 Quantifiers

When the variables in a propositional function are assigned values, the resulting statement becomes a proposition with a certain truth value. However, there is another important way, called quantification, to create a proposition from a propositional function. Quantification expresses the extent to which a predicate is true over a range of elements. In English, the words all, some, many, none, and few are used in quantifications. We will focus on two types of quantification here : universal quantification, which tells us that a predicate is true for every element under consideration, and existential quantification, which tells us that there is one or more element under

consideration for which the predicate is true. The area of logic that deals with predicates and quantifiers is called the predicate calculus.

Definition 2. The universal quantification of $P(x)$ is the statement :

" $P(x)$ for all values of x in the domain"

The notation $\forall xP(x)$ denotes the universal quantification of $P(x)$. Here $\forall$ is called the universal quantifier. We read $\forall xP(x)$ as "for all $xP(x)$ " or "for every $xP(x)$ ".

An element for which $P(x)$ is false is called a **counterexample** of $\forall xP(x)$.

Remarks :

1. The domain (or also the domain of discourse) specifies all the possible values of the variable x
2. If the domain is empty, the $\forall xP(x)$ is True for any propositional function $P(x)$ because there are no elements x in the domain for which $P(x)$ is false.

A statement $\forall xP(x)$ is false, where $P(x)$ is a propositional function, if and only if $P(x)$ is not always true when x is in the domain. One way to show that $P(x)$ is not always true when x is in the domain is to find a counterexample to the statement $\forall xP(x)$. Note that a single counterexample is all we need to establish that $\forall xP(x)$ is false.

Examples

1. Let $P(x)$ be the statement " $x + 1 > x$." What is the truth value of the quantification $\forall xP(x)$, where the domain consists of all real numbers?

Solution : Because $P(x)$ is true for all real numbers x , the quantification $\forall xP(x)$ is true.

2. Let $Q(x)$ be the statement " $x < 2$ ". What is the truth value of the quantification $\forall xQ(x)$, where the domain consists of all real numbers?

Solution : $Q(x)$ is not true for every real number x , because, for instance, $Q(3)$ is false. That is, $x = 3$ is a counterexample for the statement $\forall xQ(x)$. Thus $\forall xQ(x)$ is false.

3. Suppose that $P(x)$ is " $x^2 > 0$ ". To show that the statement $\forall xP(x)$ is false where the universe of discourse consists of all integers, we give a counterexample. We see that $x = 0$ is a counterexample because $x^2 = 0$ when $x = 0$, so that x^2 is not greater than 0 when $x = 0$.

When all the elements in the domain can be listed - say, $x_1, x_2, \dots, x_n$, it follows that the universal quantification $\forall xP(x)$ is the same as the conjunction $P(x_1) \wedge P(x_2) \wedge \dots \wedge P(x_n)$, because this conjunction is true if and only if $P(x_1), P(x_2), \dots, P(x_n)$ are all true.

Example : What is the truth value of $\forall xP(x)$, where $P(x)$ is the statement " $x^2 < 10$ " and the domain consists of the positive integers not exceeding 4?

Solution : The statement $\forall xP(x)$ is the same as the conjunction $P(1) \wedge P(2) \wedge P(3) \wedge P(4)$.

Definition 3. The existential quantification of $P(x)$ is the proposition : "There exists an element x in the domain such that $P(x)$."

We use the notation $\exists xP(x)$ for the existential quantification of $P(x)$. Here $\exists$ is called the existential quantifier.

A domain must always be specified when a statement $\exists xP(x)$ is used. Furthermore, the meaning of $\exists xP(x)$ changes when the domain changes. Without specifying the domain, the statement $\exists xP(x)$ has no meaning.

Besides the phrase "there exists" we can also express existential quantification in many other ways, such as by using the words "for some", "for at least one" or "there is". The existential quantification $\exists xP(x)$ is read as "There is an x such that $P(x)$ ", "There is at least one x such that $P(x)$ ", or "For some $xP(x)$ ".

Example :

Let $P(x)$ denote the statement " $x > 3$ ". What is the truth value of the quantification $\exists xP(x)$, where the domain consists of all real numbers?

Solution : Because " $x > 3$ " is sometimes true -for instance, when $x = 4$ - the existential quantification of $P(x)$, which is $\exists xP(x)$, is true.

Observe that the statement $\exists xP(x)$ is false if and only if there is no element x in the domain for which $P(x)$ is true. That is, $\exists xP(x)$ is false if and only if $P(x)$ is false for every element of the domain.

Example : Let $Q(x)$ denote the statement " $x = x + 1$ ". What is the truth value of the quantification $\exists xQ(x)$, where the domain consists of all real numbers?

Solution : Because $Q(x)$ is false for every real number x , the existential quantification of $Q(x)$, which is $\exists xQ(x)$, is false.

Remark :

If the domain is empty, then $\exists xQ(x)$ is false whenever $Q(x)$ is a propositional function because when the domain is empty, there can be no element x in the domain for which $Q(x)$ is true.

When all elements in the domain can be listed - say, $x_1, x_2, \dots, x_n$ - the existential quantification $\exists xP(x)$ is the same as the disjunction $P(x_1) \vee P(x_2) \vee \dots \vee P(x_n)$, because this disjunction is true if and only if at least one of $P(x_1), P(x_2), \dots, P(x_n)$ is true.

Example :

What is the truth value of $\exists xP(x)$, where $P(x)$ is the statement " $x^2 > 10$ " and the universe of discourse consists of the positive integers not exceeding 4?

Solution : Because the domain is $\{1, 2, 3, 4\}$, the proposition $\exists xP(x)$ is the same as the disjunction $P(1) \vee P(2) \vee P(3) \vee P(4)$.

Because $P(4)$, which is the statement " $4^2 > 10$ " is true, it follows that $\exists xP(x)$ is true.

Precedence of Quantifiers

The quantifiers $\forall$ and $\exists$ have higher precedence than all logical operators from propositional calculus. For example, $\forall x P(x) \vee Q(x)$ is the disjunction of $\forall x P(x)$ Or $Q(x)$. In other words, it means $(\forall x P(x)) \vee Q(x)$ rather than $\forall x (P(x) \vee Q(x))$.

Binding Variables

When a quantifier is used on the variable x , we say that this occurrence of the variable is bound. An occurrence of a variable that is not bound by a quantifier or set equal to a particular value is said to be free. All the variables that occur in a propositional function must be bound or set equal to a particular value to turn it into a proposition. This can be done using a combination of universal quantifiers, existential quantifiers, and value assignments.

The part of a logical expression to which a quantifier is applied is called the scope of this quantifier. Consequently, a variable is free if it is outside the scope of all quantifiers in the formula that specify this variable.

Examples :

- In the statement $\exists x(x + y = 1)$, the variable x is bound by the existential quantification $\exists x$, but the variable y is free because it is not bound by a quantifier and no value is assigned to this variable. This illustrates that in the statement $\exists x(x + y = 1)$, x is bound, but y is free.
- In the statement $\exists x(P(x) \wedge Q(x)) \vee \forall x R(x)$, all variables are bound. The scope of the first quantifier, $\exists x$, is the expression $P(x) \wedge Q(x)$ because $\exists x$ is applied only to $P(x) \wedge Q(x)$, and not to the rest of the statement. Similarly, the scope of the second quantifier, $\forall x$, is the expression $R(x)$. That is, the existential quantifier binds the variable x in $P(x) \wedge Q(x)$ and the universal quantifier $\forall x$ binds the variable x in $R(x)$. Observe that we could have written our statement using two different variables x and y , as $\exists x(P(x) \wedge Q(x)) \vee \forall y R(y)$, because the scopes of the two quantifiers do not overlap.

The meaning of the universal quantifier is summarized in table 1.

Statement	When True ?	When False ?
$\forall x P(x)$	$P(x)$ is true for every x	There is an x for which $P(x)$ is false.
$\exists x P(x)$	There is an x for which $P(x)$ is true	$P(x)$ is false for every x .

TABLE 1 – Meaning of the universal quantifier

4 Translating from English into Logical Expressions

Translating sentences in English (or other natural languages) into logical expressions is a crucial task in mathematics, logic programming, artificial intelligence, software engineering, and many other disciplines.

Example 1 :

Express the statement "Every student in this class has studied calculus" using predicates and quantifiers.

Solution : First, we rewrite the statement so that we can clearly identify the appropriate quantifiers to use. Doing so, we obtain : "For every student in this class, that student has studied calculus."

Next, we introduce a variable x so that our statement becomes "For every student x in this class, x has studied calculus." Continuing, we introduce $C(x)$, which is the statement " x has studied calculus."

Consequently, if the domain for x consists of the students in the class, we can translate our statement as : $\forall x C(x)$.

If we change the domain to consist of all people, we will need to express our statement as "For every person x , if person x is a student in this class then x has studied calculus."

If $S(x)$ represents the statement that "person x is in this class", we see that our statement can be expressed as : $\forall x (S(x) \rightarrow C(x))$.

Attention : Our statement cannot be expressed as $\forall x (S(x) \wedge C(x))$ because this statement says that all people are students in this class and have studied calculus!.

Example 2 :

Express the statements "Some student in this class has visited Mexico" and "Every student in this class has visited either Canada or Mexico" using predicates and quantifiers.

Solution : The statement "Some student in this class has visited Mexico" means that "There is a student in this class with the property that the student has visited Mexico." We can introduce a variable x , so that our statement becomes "There is a student x in this class having the property that x has visited Mexico." We introduce $M(x)$, which is the statement " x has visited Mexico." If the domain for x consists of the students in this class, we can translate this first statement as $\exists x M(x)$.

However, if we are interested in people other than those in this class, we look at the statement a little differently. Our statement can be expressed as "There is a

person x having the properties that x is a student in this class and x has visited Mexico.”

In this case, the domain for the variable x consists of all people. We introduce $S(x)$ to represent “ x is a student in this class”. Our solution becomes $\exists x(S(x) \wedge M(x))$ because the statement is that there is a person x who is a student in this class and who has visited Mexico.

Attention :

Our statement cannot be expressed as $\exists x(S(x) \rightarrow M(x))$, which is true when there is someone not in the class because, in that case, for such a person x , $S(x) \rightarrow M(x)$ becomes either $F \rightarrow T$ or $F \rightarrow F$, both of which are true.

Similarly consider : “For every x in this class, x has the property that x has visited Mexico or x has visited Canada.”

We let $C(x)$ be “ x has visited Canada”.

if the domain for x consists of the students in this class, this second statement can be expressed as $\exists x(C(x) \vee M(x))$.

However, if the domain for x consists of all people, our statement can be expressed as “For every person x , if x is a student in this class, then x has visited Mexico or x has visited Canada”. Thus the statement can be expressed as : $\forall x(S(x) \rightarrow (C(x) \vee M(x)))$.

5 Logical Equivalences Involving Quantifiers

Definition 4. *Statements involving predicates and quantifiers are logically equivalent if and only if they have the same truth value no matter which predicates are substituted into these statements and which domain of discourse is used for the variables in these propositional functions. We use the notation $S \equiv T$ to indicate that two statements S and T involving predicates and quantifiers are logically equivalent.*

Example : $\forall x(P(x) \wedge Q(x)) \equiv \forall xP(x) \wedge \forall xQ(x)$.

Theorem 1. De Morgan’s Law

1. $\neg \forall xP(x) \equiv \exists x \neg P(x)$
2. $\neg \exists xP(x) \equiv \forall x \neg P(x)$

Example :

Consider the negation of the statement “Every student in your class has taken a course in calculus”.

This statement can be expressed as : $\forall xP(x)$, where $P(x)$ is the statement “ x has taken a course in calculus” and the domain consists of the students in your class.

The negation of this statement is “It is not the case that every student in your class has taken a course in calculus”. This is equivalent to “There is a student in

your class who has not taken a course in calculus”. And this is simply the existential quantification of the negation of the original propositional function : $\exists x \neg P(x)$.

Consider the proposition “There is a student in this class who has taken a course in calculus”. This is the existential quantification $\exists xQ(x)$, and the domain consists to students in your class.

The negation of this statement is the proposition “It is not the case that there is a student in this class who has taken a course in calculus”. This is equivalent to “Every student in this class has not taken calculus” which is just the universal quantification of the negation of the original propositional function, or, phrased in the language of quantifiers : $\forall x \neg Q(x)$.

Table 3 summarizes DeMorgan’s Law.

Negation	Equivalent Statement	When is Negation True ?	When is False
$\neg \exists xP(x)$	$\forall x \neg P(x)$	For every x , $P(x)$ is false	There is an x for which $P(x)$ is True
$\neg \forall xP(x)$	$\exists x \neg P(x)$	There is an x for which $P(x)$ is false	$P(x)$ is True for every x

TABLE 2 – Demorgan’s Law

6 Exercises

6.1 Exercise 1 :

Express the following statement using quantifiers : “All lions are fierce”, “Some lions do not drink coffee” and “Some fierce creatures do not drink coffee”.

The domain is all the creatures.

6.2 Exercise 2 :

Express the following statement using quantifiers : “All hummingbirds are richly colored”, “No large birds live on honey”, “Birds that do not live on honey are dull in color” and “Hummingbirds are small”.

The domain consists of all birds,

7 Nested Quantifiers

Definition 5. Two quantifiers are nested if one is within the scope of the other.

Example 1 : $\forall x \exists y (x + y = 0)$. This statement means that for every real number x , it exists a real number y such that $x + y = 0$.

Example 2 : Translate into English the statement : $\forall x \forall y ((x > 0) \wedge (y < 0) \longrightarrow (xy < 0))$ where the domain for both variables consists of all real numbers.

This statement says that for every real number x and for every real number y , if $x > 0$ and $y < 0$, then $xy < 0$. That is, this statement says that for real numbers x and y , if x is positive and y is negative, then xy is negative. This can be stated more succinctly as "The product of a positive real number and a negative real number is always a negative real number."

THINKING OF QUANTIFICATION AS LOOPS : In working with quantifications of more than one variable, it is sometimes helpful to think in terms of nested loops :

1. To see whether $\forall x \forall y P(x, y)$ is true, we loop through the values for x , and for each x we loop through the values for y . If we find that $P(x, y)$ is true for all values for x and y , we have determined that $\forall x \forall y P(x, y)$ is true. If we ever hit a value x for which we hit a value y for which $P(x, y)$ is false, we have shown that $\forall x \forall y P(x, y)$ is false.
2. To determine whether $\forall x \exists y P(x, y)$ is true, we loop through the values for x . For each x we loop through the values for y until we find a y for which $P(x, y)$ is true. If for every x we hit such a y , then $\forall x \exists y P(x, y)$ is true; if for some x we never hit such a y , then $\forall x \exists y P(x, y)$ is false.
3. To see whether $\exists x \forall y P(x, y)$ is true, we loop through the values for x until we find an x for which $P(x, y)$ is always true when we loop through all values for y . Once we find such an x , we know that $\exists x \forall y P(x, y)$ is true. If we never hit such an x , then we know that $\exists x \forall y P(x, y)$ is false.
4. To see whether $\exists x \exists y P(x, y)$ is true, we loop through the values for x , where for each x we loop through the values for y until we hit an x for which we hit a y for which $P(x, y)$ is true. The statement $\exists x \exists y P(x, y)$ is false only if we never hit an x for which we hit a y such that $P(x, y)$ is true.

7.1 Orders of Quantifiers

Many mathematical statements involve multiple quantifications of propositional functions involving more than one variable. It is important to note that the order of

Statement	When True ?	When False ?
$\forall x \forall y P(x, y)$ or $\forall y \forall x P(x, y)$	$P(x, y)$ is true for every pair x, y .	There is a pair x, y for which $P(x, y)$ is false.
$\forall x \exists y P(x, y)$	For every x , there is y for which $P(x, y)$ is true	There is x for which $P(x, y)$ is false for every y .
$\exists x \forall y P(x, y)$	There is x for which $P(x, y)$ is True for every y	For every x , there is y for which $P(x, y)$ is false
$\exists x \exists y P(x, y)$ or $\exists y \exists x P(x, y)$	There is a pair x, y for which $P(x, y)$ is true	$P(x, y)$ is false for every pair x, y .

TABLE 3 – Quantifications of Two Variables.

the quantifiers is important, unless all the quantifiers are universal quantifiers or all are existential quantifiers.

Example 1 : Let $P(x, y)$ be the statement " $x + y = y + x$ ". What are the truth values of the quantifications $\forall x \forall y P(x, y)$ and $\forall y \forall x P(x, y)$ where the domain for all variables consists of all real numbers ?

The quantification $\forall x \forall y P(x, y)$ denotes the proposition "For all real numbers x , for all real numbers y , $x + y = y + x$. Because $P(x, y)$ is true for all real numbers x and y (it is the commutative law for addition).

Note that the statement $\forall y \forall x P(x, y)$ says "For all real numbers y , for all real numbers x , $x + y = y + x$." This has the same meaning as the statement "For all real numbers x , for all real numbers y , $x + y = y + x$." That is, $\forall x \forall y P(x, y)$ and $\forall y \forall x P(x, y)$ have the same meaning.

Example 2 : Let $Q(x, y)$ denote " $x + y = 0$ ". What are the truth values of the quantifications $\exists y \forall x Q(x, y)$ and $\forall x \exists y Q(x, y)$, where the domain for all variables consists of all real numbers ?

The quantification $\exists y \forall x Q(x, y)$ denotes the proposition "There is a real number y such that for every real number x , $Q(x, y)$. No matter what value of y is chosen, there is only one value of x for which $x + y = 0$. Because there is no real number y such that $x + y = 0$ for all real numbers x , the statement $\exists y \forall x Q(x, y)$ is false.

The quantification $\forall x \exists y Q(x, y)$ denotes the proposition "For every real number x there is a real number y such that $Q(x, y)$." Given a real number x , there is a real number y such that $x + y = 0$; namely, $y = -x$. Hence, the statement $\forall x \exists y Q(x, y)$ is true.

7.2 Negating Nested Quantifiers

Statements involving nested quantifiers can be negated by successively applying the rules for negating statements involving a single quantifier (De Morgan's law).

Example 1 : Express the negation of the statement $\forall x \exists y (xy = 1)$ so that no negation precedes a quantifier. $\neg \forall x \exists y (xy = 1) \equiv \exists x \neg \exists y (xy = 1) \equiv \exists x \forall y \neg (xy = 1)$ thus the statement is equivalent to $\exists x \forall y (xy \neq 1)$

8 Exercises :

Exercise 1 :

Let $T(x, y)$: "x eats y", $w(X)$: "x is wise", $H(x)$: "x is strong", $C(X)$: "y is juicy", $R(y)$: "y is red". The domain for X the students = Jack, Mike, Harry and Y domain are the fruits = Apple, Orange, Banana. Assume Jack eats Apple and Banana. Mike eats all fruit and Harry eat oranges and apples. Also assume Mike and Harry are wise and apple and orange are red. Convert from English to symbolic and vice versa and find the truth for each.

1. All students eat all fruits.

2. Some students eat some fruits.

3. All students eat some fruits.

4. Some students eat all fruits.

5. Jack eats all fruits.

6. Some students eat orange.

7. Harry does not eat banana.

8. All fruits are eaten by all students.

9. Some fruits are eaten by some students.

10. All fruits are eaten by some students.

11. Some fruits are eaten by all students.

12. All wise student eat all red fruits.

13. All non red fruits are eaten by some students.

14. All neither wise nor strong students do not eat banana.

15. For all student x and all fruits y : if x is wise and y not red, then x does not eat y.

16. If a student either strong or not wise , then he eats some red juicy fruits.

17. There exists a not wise or strong student that eat all fruits.

18. Some fruits are eaten by all wise and strong students.

19. some wise students do not eat some not juicy fruits.

20. some juicy or not red fruits are eaten by all not strong students.

21. No wise student do not eat all red fruits.

22. Not all students are wise or strong.

23. No red fruits is not juicy.

24. Not all strong wise students eat Banana.

25. No students do not eat all red or juicy fruits.

26. $\forall x, \forall y : W(x) \wedge \neg R(y) \longrightarrow T(x, y)$.

27. $\forall x, \exists y : \neg W(x) \longrightarrow \neg T(x, y) \wedge C(y)$.

28. $\neg \forall x, \exists y : \neg W(x) \wedge \neg H(x) \longrightarrow T(x, y)$.

29. $\exists x, \exists y : H(x) \wedge \neg C(y) \wedge T(x, y)$.

30. $\exists y, \forall x : R(y) \vee \neg C(y) \wedge [W(x) \longrightarrow T(x, y)]$.

31. $\neg \forall y : \neg C(x) \longrightarrow R(y)$.

32. $\neg \exists x : W(x) \wedge H(x)$.

Exercise 2 :

$P(x, y)$: "x bites y" , $B(x)$: "x is black" , $F(x)$ "x is furry". $G(x)$: "x is grumpy".
 $C(x)$: "x is clumsy". the domain for x and y are all cats.

Convert from symbolic to English and vice versa.

1. the black Catoo bites the clumsy Fifi.

2. Fifi bites all furry cats.

3. All grumpy cats bite Catoo but not Fifi.

4. Some black cats do not bite clumsy Fifi.

5. All non black grumpy cats bite all cats.

6. All non black cats bite all grumpy cats.

7. All grumpy cats are bitten by all non black cats.

8. All cats are neither non black nor bite Catoo.
9. All cats are grumpy and bite all cats.
10. All cats are not black or bite all furry cats .
11. All cats bite some non clumsy cats.
12. All grumpy cats bite some black cats.
13. All cats do not bite some black cats.
14. Some cats bite the clumsy black Catoo .
15. Some furry cats are not bitten by the non furry Fifi.
16. Some not furry black cats bite some cats.
17. There is a non clumsy cat such that it bites some black cats.
18. Some black grumpy cats bite all cats.
19. Some neither grumpy nor clumsy cats bite all either black or furry cats.
20. All neither grumpy nor clumsy cats are bitten by some either black or furry cats.
21. Some neither grumpy nor clumsy cats do not bite all cats.
22. $\forall y : P(Fifi, y) \vee B(y)$.
23. $\forall y : B() \wedge \neg C(y)$.
24. $P(Fifi, Catoo) \longrightarrow G(Fifi) \vee B(Catoo)$.
25. $\exists x : P(x, Catoo) \wedge B(x)$.
26. $\forall x, \forall y : \neg B(x) \wedge \neg P(x, y)$.
27. $\forall x, \forall y : \neg B(x) \longrightarrow \neg P(x, y)$.
28. $\forall x, \forall y : \neg P(x, y) \longrightarrow \neg B(x)$.
29. $\forall x, \forall y : \neg B(y) \longrightarrow \neg P(x, y)$.

$$30. \forall y, \forall x : \neg B(y) \longrightarrow \neg P(x, y).$$

$$31. \forall x, \exists y : G(x) \wedge C(x) \longrightarrow P(x, y) \wedge F(y).$$

$$32. \forall x, \exists y : \neg G(x) \vee C(x) \longrightarrow P(x, y).$$

$$33. \forall y, \exists x : \neg F(y) \longrightarrow P(x, y) \wedge F(x).$$

$$34. \forall x, \exists y : P(x, y) \wedge [F(y) \vee \neg B(y)].$$

$$35. \exists x, \forall y : G(y) \longrightarrow P(x, y).$$

$$36. \exists x, \forall y : B(x) \wedge [G(y) \longrightarrow P(x, y)].$$

$$37. \exists y, \forall x : B(y) \wedge P(x, y).$$

$$38. \exists y, \exists x : B(y) \wedge P(x, y) \wedge G(x).$$

$$39. \exists x, \exists y : B(x) \wedge P(x, y) \wedge G(y).$$

$$40. \neg \forall x : B(x) \vee \neg F(x).$$

$$41. \neg \forall x : B(x) \longrightarrow \neg F(x).$$

$$42. \neg \forall x \forall y : B(x) \wedge \neg G(y) \longrightarrow P(x, y).$$

$$43. \neg \forall y \exists x : \neg B(y) \longrightarrow P(x, y) \wedge C(x).$$

$$44. \neg \exists x \forall y : \neg G(x) \wedge [B(y) \longrightarrow P(x, y)].$$

$$45. \neg \exists x \exists y : C(x) \wedge \neg B(y) \wedge \neg P(x, y).$$

$$46. \neg \exists y : \neg B(y) \wedge \neg P(Catoo, y) \wedge \neg F(Catoo).$$

$$47. \neg \exists y \exists x : C(y) \wedge P(x, y).$$

Exercise 3 :

Find if the statement true or false. Justify your answer.

$$1. \exists x \in \mathbb{Z}, \forall y \in \mathbb{Z}^+ : xy < 0.$$

$$2. \forall x \in \mathbb{Z}, \exists y \in \mathbb{Z}^+ : xy < 0$$

$$3. \forall y \in \mathbb{R}, \exists x \in \mathbb{R} : y = 2x + 1$$

$$4. \exists x \in \mathbb{R}, \exists y \in \mathbb{R} : (x - y = -9) \wedge (2x + y = 3).$$

$$5. \forall y \in \mathbf{Z}^+, \exists x \in \mathbf{Z}^+ : y = \sqrt{x+2}.$$

$$6. \exists a \in Z, \forall b \in Z : a + b = 2.$$

$$7. \exists x \in R : \sqrt{\frac{x}{\sqrt{2}}} \text{ is rational but not integer.}$$

$$8. \forall p \in N : \text{if } p \text{ is divisible by } 5, \text{ then } 2^p - 1 \text{ is prime.}$$

$$9. \forall n \in Z^+, \exists t \in Q^+ : t^2 - n - 1 = 0.$$

$$10. \exists a, b \in Z : \text{if } a > b \text{ then } a^2 + b < 0.$$

$$11. \exists x, y \in Z : x - y = y.$$

$$12. \forall x, y \in Z : x - y = y.$$

$$13. \forall x \in Z, \exists y \in Z : x - y = y.$$

$$14. \exists x \in R, \forall y \in R : x - y = y.$$

$$15. \exists x, y \in Z : \frac{x}{y} = x + y .$$

$$16. \forall x, y \in Z : x^2 + y^2 > 0.$$

$$17. \forall x \in Z, \exists y \in Z : \frac{x^2 - 2y}{y^2 + 1} = 1.$$

$$18. \exists x \in Z, \forall y \in R : \sqrt{xy^2} = 0.$$

$$19. \forall x \in Z, \forall y \in Z : xy = \sqrt{x}\sqrt{y}.$$

$$20. \forall x \in Z, \exists y \in Z : xy = 1.$$

$$21. \exists x \in Z^+, \exists y \in Z^+ : x\sqrt{y} = 1.$$

$$22. \exists x \in A : x^2 - \sqrt{x} = 0 \text{ where } A = 0, 1, 2.$$

$$23. \forall x \in R, \forall y \in R : x^2 < y + 1.$$

$$24. \forall x \in Z, \exists y \in Z^- : x^2 - y^2 = 9.$$

$$25. \exists x \in R, \forall y \in R : y < 5 \longrightarrow y^2 - x^2 \geq 0.$$

$$26. \forall x \in Z : (x^2 + 2x - 3 = 0) .$$

$$27. \exists x \in Z \exists k \in Z : (x = 2k + 1) \wedge (x^2 - 2x + 5 \leq 0).$$

$$28. \exists m \in R, \forall x \in Z^+ : \frac{x}{x+1} < m.$$

$$29. \forall x \in Z, \exists y \in Z : (x^2 - y^2 = 4x - 4).$$